

BI4016 Research Project Thesis
2025/2026 School of Biological Sciences

**Synergistic and independent mechanisms of generating
plasticity in response to benzo[a]pyrene exposure in
Caenorhabditis elegans and *Gadus morhua***

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Work completed in partial fulfillment of the requirements for the
degree BSc Environmental Science

Word count excluding figures and references: 5308

Abstract

Background: Before selection pressures cause adaptation, organisms counter environmental stressors via phenotypic plasticity. Differential gene expression (DGE) and alternative splicing (AS) are mechanisms that allow organisms to modify their transcriptome and proteome in response to challenges. Benzo[a]pyrene (BaP) is a ubiquitous environmental carcinogen that elicits plasticity in exposed individuals in a dose responsive manner. Despite evidence that DGE and AS play important roles in generating phenotypic novelty, it is unclear which genes AS takes place in and the extent of AS in dose responses.

Research Question: This study evaluates if DGE and AS occur in the same genes, if rates of DGE and AS increase as BaP exposure increases, and if effects of exposure on DGE and AS can be modelled using dose as a numerical predictor.

Methods: Publicly available global transcriptomic data from *Caenorhabditis elegans* and *Gadus morhua* exposed to 0, 5, 20 and 0, 0.01, and 1 uM BaP, respectively, was analyzed for DGE and AS. Functional enrichment was performed with DAVID to examine functional relatedness of genes where DGE and AS took place.

Results: 8% of genes detected in *C. elegans*' DGE and AS analyses overlapped, exhibiting synergism, but no genes overlapped in *G. morhua*. Rates of DGE and AS increased following the dose gradient in *C. elegans* only. Modelling BaP exposure as a numerical predictor yielded 7x and 3x as many genes as detected in traditional pairwise contrast analyses in *C. elegans* and *G. morhua*, respectively.

Discussion: Evidence suggests that AS was an independent mechanism to generate plasticity to DGE in most genes, but that it exhibits synergism in the most important gene that translates a BaP-metabolizing enzyme (*cyp35a1*). It is shown that AS increases as BaP dose increases. Linear modelling of dose effects on transcription may not be complex enough to capture the complexity of transcriptional regulation unless experimental designs are constructed for numerical prediction.

Background

Organisms must counter environmental challenges at the molecular level to maintain homeostasis. Short term phenotypic plasticity is a temporary shield to potentially detrimental exposures that takes place before long-term adaptation occurs. One source of phenotypic plasticity is gene expression. Differential gene expression (DGE) modifies the transcription rate of particular genes from a ground state to an advantageous ratio of transcripts (Oomen & Hutchings, 2017). While phenotypic plasticity is generated by gene-level expression changes, it is also generated post-transcriptionally (Bush et al., 2017). Alternative splicing (AS) produces transcript isoforms that differ in the spliced exons included in the final transcript, the exons' order, and whether intronic sequences are retained (Verta & Jacobs, 2022). The resulting mature mRNA may then be translated into functional proteins that allow the organism to respond to the environmental challenge. DGE and AS are distinct mechanisms that mediate an organism's phenotype by changing which genes and what gene product isoforms are transcribed.

The advent of accessible, high-throughput RNA sequencing (HTP RNAseq) has presented a unique opportunity to evaluate how gene expression and splicing operate in ecological contexts. Previously, gene expression was implicated in plastic responses throughout the animal kingdom: in the protection against cold snaps in lizards (Campbell-Staton et al., 2017), in the ecotyping of genetically homogenous southern pygmy perch populations (Brauer et al., 2017), and in the opsin-engineering across teleosts (Bertinetti & Torres-Dowdall, 2025). Conversely, alternative splicing was shown to support single-gene, morphological adaptations: in freshwater threespine sticklebacks (Howes et al., 2017) and speciated East African cichlids (Terai et al., 2003). Now, researchers are evaluating how DGE and AS contribute to short-term phenotypic plasticity and whether the 2 are synergistic processes occurring in the same genes, or independent processes, occurring in different genes.

Ecological studies have demonstrated that a wide range of stressors trigger synergistic and independent phenotypic plasticity. It was shown that genes involved in DGE and AS differed in pairs of Arctic char across 3 ecotypes (Jacobs & Elmer,

2021). While alternatively spliced genes in swimming muscle were enriched for muscle development and functioning, differentially expressed genes were enriched for metabolic and developmental processes. Similarly, Steward et al (2022) found that induced climatic variation led to alternative splicing in genes that were lowly differentially expressed in *Bicyclus* butterflies. On the other hand, researchers have also shown DGE and AS are synergistic. For example, DGE and AS took place in a similar number and type of receptor genes, including the Down syndrome cell adhesion molecule gene in the *Bombus terrestris*-*Crithidia bombi* host-parasite system (Riddell et al., 2014). Vaidya et al (2025) found that differentially expressed and alternatively spliced genes overlapped in heat shock proteins of intertidal copepods, but did not overlap in cytoskeleton, metabolism, or chitin genes. Despite improvements in tools to measure how organisms modify phenotypes in response to stress, the relationship between differential gene expression and alternative splicing is still unclear, especially in ecological contexts.

Dose response is an established concept in toxicology that characterizes how gradients of chemical exposures result in dose-dependent effects on the target organism (Tsatsakis et al., 2018). In the context of molecular ecology, an organism exposed to a low amount of an effector may exhibit more DGE compared with a control. Further, a greater exposure may lead to more DGE than both the control and low treatment. Benzo[a]pyrene (BaP) is a 5 ring polycyclic aromatic hydrocarbon (PAH) that is ubiquitous in the environment and forms during the pyrolysis of organic matter (Phillips, 1999). BaP is a group 1 carcinogen due to evidence that it causes DNA adduction in humans and animals (IARC, 2012) and exhibits dose-dependent health outcomes (Tartaglione et al., 2023). PAHs are known to be bioactivated by genes in the *CYP* superfamily in eukaryotes (Shimada & Fujii-Kuriyama, 2004; Harris et al., 2020; Zhu et al., 2024), and the *cyp1* family is known to bioactivate BaP in humans specifically. *Caenorhabditis elegans* lacks *cyp1* genes but it is believed that *cyp1*'s role in human BaP metabolism is replaced by other members including *cyp35* in *C. elegans* (Leung et al., 2010; Harlow et al., 2018; Lim et al., 2022).

Benzo[a]pyrene is frequently used as a representative of PAHs and environmental pollutants. Previous research has shown that BaP elicits dose-dependent effects on gene expression. When *Caenorhabditis elegans* and *Gadus morhua* precision-cut liver slices (Atlantic cod PCLS) were exposed to 0, 5, 20 and 0, 0.01, and 1 μ M

benzo[a]pyrene, respectively, researchers found an increase in the number of differentially expressed genes (DEGs) following the dose gradient (Yadatie et al., 2018; Abbass et al., 2021). The low exposure groups in *C. elegans* and *G. morhua* yielded 136 and 1 DEG, respectively. The higher exposure groups yielded 238 and 13 DEGs. Fewer studies have bridged the toxicological principle of dose response from gene expression to splicing in ecological contexts. It is unknown if rates of alternative splicing correlate with changes to rates of DGE.

Deposited HTP RNAseq data from *C. elegans* and *G. morhua* is used to investigate if DGE and AS are synergistic processes, occurring in the same genes, or independent processes, occurring in different genes, when exposed to increasing concentrations of BaP. Secondly, the study investigates whether higher concentrations of BaP increases the rate of DGE and AS. Thirdly, the study examines if modelling BaP exposure as a predictor of gene expression and alternative splicing explains per-unit, dose-dependent effects on transcriptional modification in supplement to classical pairwise contrast analyses. Finally, this comparative study benefits from using 2 reporter species to underline how DGE and AS operate in species that use different pathways to metabolize benzo[a]pyrene. It is hypothesized that there is little to no overlap between DGE and AS, that rates of AS will increase alongside DGE in an effort to cope with greater chemical stress, and that dosimetry modelling will predict low-magnitude gene expression and splicing.

Methodology

Data Acquisition

RNA sequencing data was obtained from published studies that examined DGE in response to BaP in *C. elegans* (Abbas et al., 2021; NCBI BioProject: PRJNA638820) and *G. morhua* (Yadatie et al., 2018; NCBI BioProject: PRJNA418681), respectively.

In the *C. elegans* study, OP50 *Escherichia coli* were mixed with BaP dissolved in dimethyl sulfoxide (DMSO) and were incubated with at least 7000 N2 Bristol worms per replicate. Treatments included a DMSO control group (n=3), a 5 uM BaP-exposed group (n=3), and a 20 uM BaP-exposed group (n=3). Following 48

hours of exposure, worms were frozen and prepared for RNA extraction. RNA was sequenced on an Illumina HiSeq 2500 sequencer. Reads were 50 bases in length, unpaired, and unstranded.

In the *G. morhua* study, BaP dissolved in DMSO was applied to cultured juvenile precision-cut liver slices. Treatments included a DMSO control group (n=8), a 0.01 uM BaP-exposed group (n=7), and a 1 uM BaP-exposed group (n=7). The study used a nested design. Replicates were paired within a single fish's liver slices. Following 48 hours of exposure, slices were frozen and prepared for RNA extraction. RNA was sequenced on an Illumina HiSeq 4000 sequencer. Reads were 75 base pairs in length, paired, and stranded.

Statistical Analyses

Reads were retrieved and analyzed on the University of Aberdeen MacLeod high performance computing Linux command line. Quality control was performed using FastQC (0.12.1; Andrews, 2010) and MultiQC (1.27.1; Ewels et al., 2016). Reads below quality scores of 20 were trimmed with TrimGalore (0.6.10; Krueger, 2015). Following assurance of trimmed read quality using FastQC and MultiQC again, reads were aligned against the *C. elegans* (WBcel235, NCBI BioProject: PRJNA13758) and *G. morhua* (gadMor3.0, NCBI BioProject: PRJEB33455) reference genomes with HISAT2 (2.2.1; Kim et al., 2019). Aligned reads were counted with featureCounts (subread 2.0.6; Liao et al., 2013; Liao et al., 2014) at the gene- and transcript-level using gff3 annotations, and at the exon-level using custom flattened exons. Reads that matched to multiple annotations were fractionally counted in both organisms. In *G. morhua*, 2 additional optimizations of the featureCounts command were included because reads were paired and stranded. Only reads that aligned to both ends of transcripts were counted, and no read pairs where one pair matched a different chromosome to the other were counted. Details about which options were included can be found in the Data & Code Accessibility section.

Counted reads were analyzed for DGE between treatments in RStudio (R Core Team, 2025) using DESeq2 (1.48.2; Love et al., 2014) and the gene-level read counts. Transcript- and exon-level reads were analyzed for differential transcript usage (DTU) and differential exon usage (DEU) between treatments, respectively,

using DEXSeq (1.54.1; Anders et al., 2012). The nested design employed in the *G. morhua* experimental design was accounted for by considering the individual that the PCLS samples came from in DGE and DTU analyses, but was not accounted for in the DEU analyses due to excessive model complexity. Categorical models were fitted to examine DGE, DTU, and DEU between the controls and each treatment. Additionally, linear models were fitted that used BaP exposure as a numerical predictor in order to directly estimate per unit BaP effects on gene expression and splicing (hereafter linear models).

To decrease false positives, gene-level reads were filtered and were only included if they exhibited $\pm 1 \log_2$ fold changes in comparison with the control (i.e. half or twice as many absolute counts). Additionally, genes were only presented if multiple testing-adjusted p-values ≤ 0.05 . In transcript- and exon-level contrast models, filtering used q-values ≤ 0.05 . In linear models, the only filtering used was the adjusted p- or q-values.

To demonstrate overlaps between differentially expressed and alternatively spliced genes detected across model types at the gene-, transcript-, and exon-levels, UpSet plots were created using the UpSetR package (Conway et al., 2017). Principal component analysis (PCA) was performed using variance stabilizing counts with DESeq2 to gauge variance within treatments. Volcano plots were created with the EnhancedVolcano package to visualize DEGs (Blighe et al., 2025). To investigate whether the detected genes in DGE, DTU, and DEU analyses were functionally related, functional enrichment was performed. Genes were submitted to DAVID using its default settings (Stringency: Medium & EASE: 1; Huang et al., 2008; Sherman et al., 2022; version December 2021: Accessed 7 November, 2025). DAVID assigned GeneOntology (GO) biological process terms and Kyoto Encyclopedia of Genes and Genome (KEGG) pathways to each gene and tallied how many genes matched to each GO term or KEGG pathway. DAVID returned tables containing functionally clustered term and pathway tallies, and the results of statistical tests examining whether the functional cluster and term and pathway were overrepresented compared to what would be detected by chance. This included Benjamini-Hochberg false discovery rates (FDR, a multiple testing-adjusted p-value).

The top 3 terms and top pathway in each cluster, defined by lowest $FDR \leq 0.05$, in the top 3 clusters, ranked by highest modified Fisher's exact score, were selected for presentation.

Results

The untrimmed reads contained an average of 32.94 ± 2.80 million bases for *C. elegans* and 59.79 ± 8.26 million bases for *G. morhua*. Following trimming, *C. elegans* reads were an average of 32.88 ± 2.77 million bases long and *G. morhua* reads were an average of 59.22 ± 8.27 million bases long. The results of trimming and quality control measures suggest that the reads were of high quality. The average alignment of the 9 *C. elegans* samples to the reference genome was 92%. However, only 67% of the 3rd sample in the 20 uM treatment was aligned. The average assignment to gene- and transcript-level annotations, and flattened exons, was between 83% and 93%, although the final 20 uM sample was consistently below 70% for each count level.

The average alignment of the 22 paired *G. morhua* samples to the reference genome was 93% and all samples mapped similarly. The average assignment to gene- and transcript-level annotations, and flattened exons, was lower: between 61% and 72%. This may have been due to *G. morhua* having less comprehensive annotations compared with the model organism *C. elegans*.

Principal components analysis (PCA) was performed to gauge variation within treatments. *C. elegans* reads were mostly clustered by treatment (Figure 1A). However, the sample with lower genome alignment rates and lower assignment rates had more variation in its 2nd principal component (top 20 uM sample, Figure 1A). 19 upregulated DEGs were detected in the low contrast, 49 upregulated and 5 downregulated DEGs in the high contrast, and 310 upregulated and 489 downregulated DEGs in the linear model (Table 1). The abundance of genes detected in the linear model suggests that many genes were subtly up- and down-regulated in *C. elegans*, but not at sufficient magnitude to meet the minimum fold change stipulation to be detected in the contrast models.

G. morhua reads exhibited no clustering by treatment (Figure 1B). Rather, reads were clustered by individual. 5 upregulated DEGs were detected in the high exposure model, and 13 upregulated and 2 downregulated DEGs in the linear model (Table 1).

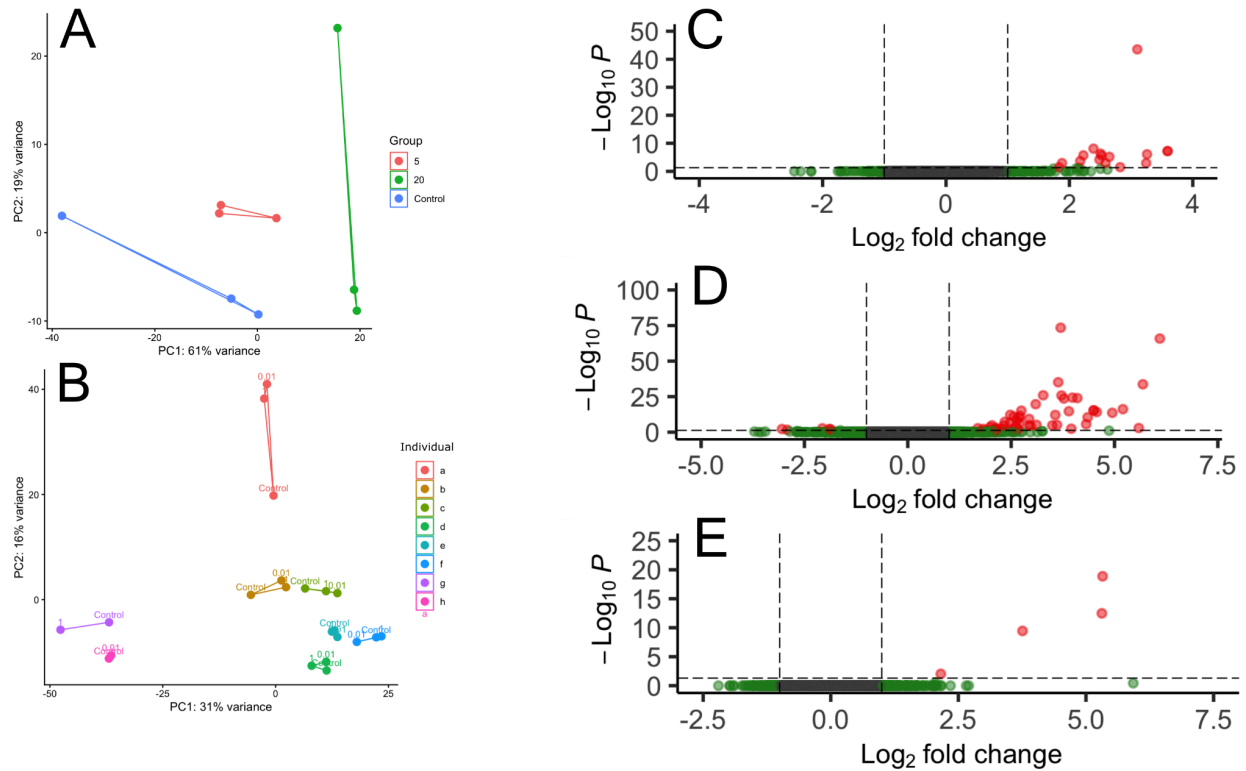


Figure 1: PCA was performed using the variance stabilizing transformation function in DESeq2 on quantified gene count data (Love et al., 2014). Analysis used the top 500 genes with greatest variance by default. Volcano plots were used to visualize $\log_2$ fold change (FC) and $-\log_{10}$ adjusted p-values in DEGs. Note that scales differ between contrasts. In (A) *C. elegans*, 80% of variance is explained by PC1 and PC2, and hulls were applied to show variance within treatments. In (B) *G. morhua*, only 47% of variance is explained by PC1 and PC2. Samples are colored according to the fish that liver slices were cut from, evidencing that intratreatment variability was greater than interindividual variation. Hulls show variance within individuals. (C) 5 uM *C. elegans* contrast. 29 upregulated genes with $\log_2$ fold changes > 1 (2x absolute expression changes) and $-\log_{10}$ adjusted p-values > 1.3 (0.05 on non-log scale) were detected. Only 1 gene was omitted (*irg5*) which had a FC of 6.6 and p-value of 175.1 to better visualize the other DEGs. (D) 20 uM *C. elegans* contrast. Most genes were upregulated and had lower adjusted p-values and higher fold changes than the 5 uM contrast. *Irg5* was omitted again ($\log_2$ FC of 8 and $-\log_{10}$ adjusted p-value of 279.1). (E) 1 uM *G. morhua* contrast. Adjusted p-values were relatively lower than in *C. elegans* contrasts. 1 gene (*cyp1a*) was omitted ($\log_2$ FC of 5.9 and p-value of 26).

Table 1: Count of genes that evidenced differential gene expression, differential transcript usage, and differential exon usage in *Caenorhabditis elegans* and *Gadus morhua*. Low exposures were 5 uM benzo[a]pyrene for *C. elegans* and 0.01 uM for *G. morhua*. High exposures were 20 uM BaP for *C. elegans* and 1 uM for *G. morhua*.

	<i>C. elegans</i>	<i>G. morhua</i>
DGE		
Low Exposure	19	0
High Exposure	54	5
Linear Model	799	15
DTU		
Low Exposure	0	0
High Exposure	0	0
Linear Model	2	0
DEU		
Low Exposure	19	0
High Exposure	29	0
Linear Model	93	1
Total	1015	21

Because the linear model predicted different DEGs to the 1 uM contrast, there may be some genes which were differentially expressed in the 0.01 and 1 uM contrasts which did not meet the minimum fold change. However, the linear model evidenced that statistically significant up- and down-regulation of low-magnitude genes took place.

There were only 2 genes that underwent differential transcript usage in the linear model in *C. elegans*. DTU was not detected in *G. morhua*. However, DEU took place in 19 genes in the low contrast, 29 in the high contrast, and 93 in the linear model. Despite little DTU, there is evidence that alternative splicing took place in many genes in *C. elegans*. In *G. morhua*, DEU was detected in 1 gene in the linear model. Because DEU was not detected in the 1 uM contrast, this suggests that low levels of alternative splicing may have taken place in the 0.01 uM contrast that contributed to the linear model's prediction (Table 1).

Fold changes in *C. elegans* DEGs were greater in the 20 uM contrast than the 5 uM contrast, and adjusted p-values were similarly lower. Expression was positively regulated in most genes as dose increased. Because no DEGs were detected in the *G. morhua* 0.01 uM contrast, it is less clear how dose affected expression. However,

the DEGs that were detected in the 1 uM contrast exhibited high fold changes relative to their adjusted p-values (Figure 1E).

An intersection analysis on genes detected by gene-, transcript-, and exon-level analyses was performed for *C. elegans* (Figure 2). Most overlap in genes took place within the gene- and exon-level, not across count levels. 8% of intersections (7 genes) took place across levels and are highlighted in light and dark blue. The light blue intersections represent synergism between differential gene expression and alternative splicing. The intersections between the linear DGE and DEU models (*Y37E3.8*, *rmh1*, *emb9*, *vap2*), the linear DTU and DEU models (*E01G4.3*), and the linear DGE and 5 uM DEU contrast (*alg1*), yielded a total of 6 genes without a specific role in immune response or xenobiotic metabolism. There was 1 gene that intersected all DGE analyses and the DEU contrasts which codes for a detoxification enzyme (*cyp35A1*).

Fewer genes were detected in gene-, transcript-, and exon-level analyses in *G. morhua* than *C. elegans*. There were 4 intersections between the 1 uM DGE contrast and linear DGE model which suggests that the 1 alternatively spliced gene (*C1orf43*) was independent to the detected DEGs (Figure 3). Related genes were detected in *C. elegans* that are involved in xenobiotic metabolism processes to *cyp1a* and *cyp1b1*. The arginine vasopressin receptor 1A is expressed in many tissues including the liver, and is enriched for a variety of biological processes which differ by species and sex.

Next, patterns in the functions of differentially expressed and alternatively spliced genes were examined. Genes detected in DGE and DEU contrasts were converted from WBGene IDs to their gene symbols and summarized (Table 2). The 2 genes detected in the DTU analysis (*E01G4.3* & *F17C11.4*) did not match to gene families with at least 1 other member. More cytochrome P450s and glucuronosyltransferases (*UGTs*) were detected in the high exposure contrast compared to the low exposure at the gene-level. The *CYP* superfamily codes for enzymes that use heme as a cofactor to oxidize xenobiotics. *UGTs* are similarly crucial for phase II metabolism because they conjugate glucuronic acid to non-polar xenobiotics, improving their excretion.

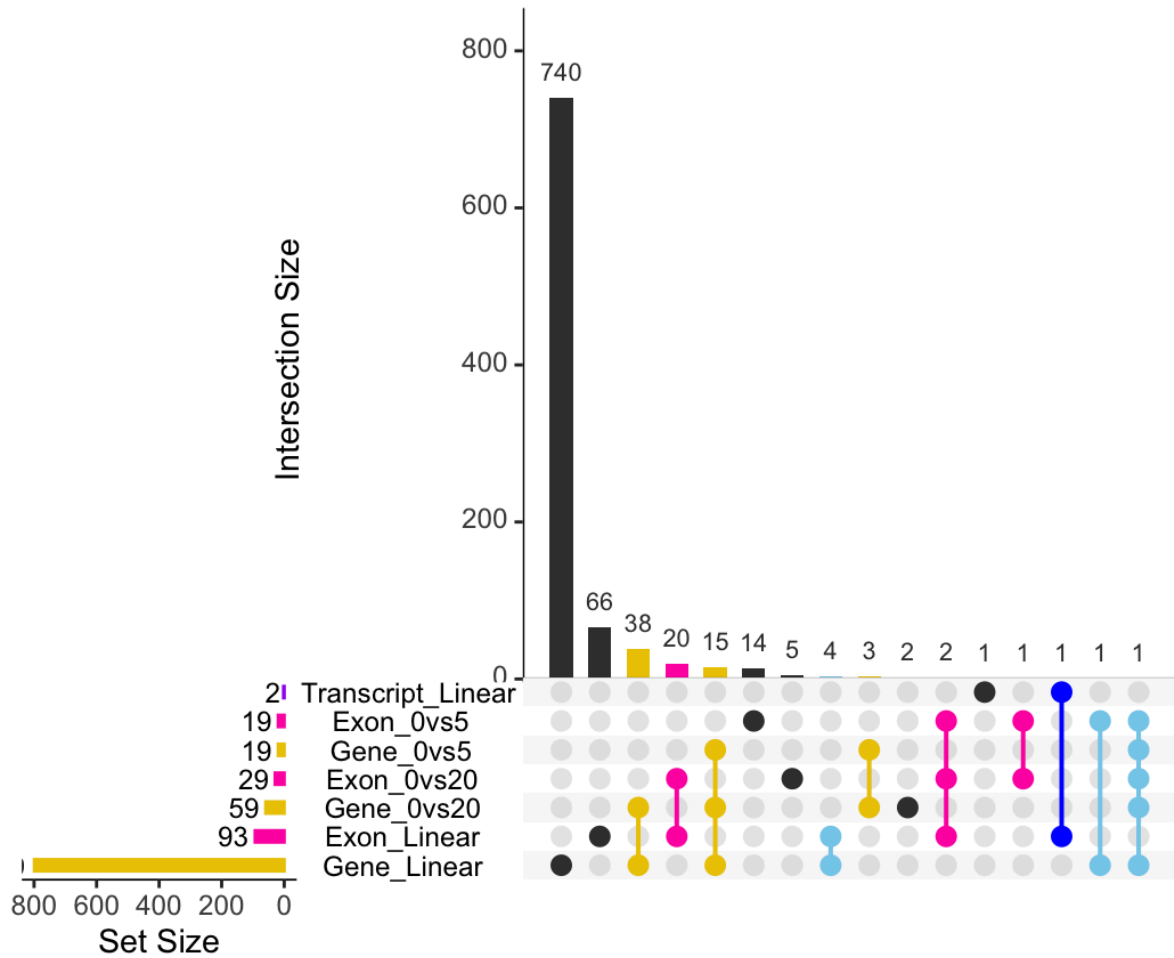


Figure 2: Intersection plot for differentially expressed genes and alternatively spliced differential transcript and exon usage genes in *C. elegans*. Gene-level set intersections are highlighted in gold. The transcript-level set is highlighted in purple. Exon-level sets are highlighted in pink. Intersections between sets of the same feature-level (i.e. gene-gene or exon-exon) are highlighted in their set color. Intersections across feature-level that suggest that DGE and AS occur in the same genes are highlighted in sky blue. The 1 gene that intersects the transcript-level model and linear exon-level model is highlighted in dark blue. Points that do not intersect other sets represent the unique genes in that model.

Additionally, alternative splicing was detected in both treatments in genes related to muscle development: *let2*, *unc15*, *unc52*, *unc54*, and *ttn1*.

In *G. morhua*, there was less patterning in detected genes. Aside from the 2 *cyp* genes detected in the 1 uM gene-level contrast and linear model (Figure 3), there were no gene families with more than 1 member represented.

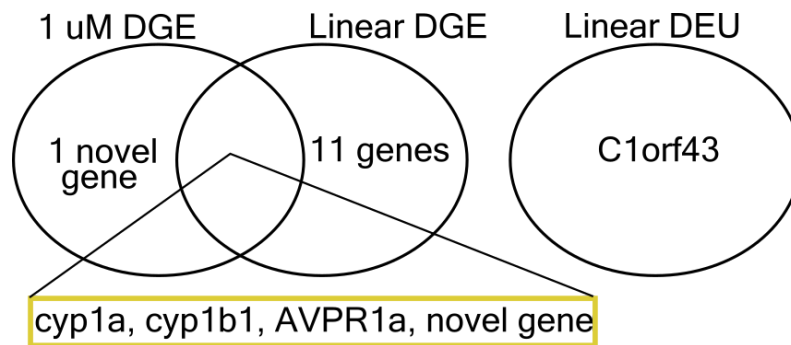


Figure 3: Intersections in differentially expressed genes detected in *Gadus morhua* analyses. 4 genes overlap between the gene-level 1 uM contrast and linear model and are shown in the gold box. 1 gene was alternatively spliced and was detected in the linear differential exon usage model but did not overlap with gene-level results.

Table 2: Gene family summary for *C. elegans* differentially expressed genes and differential exon usage (AS) genes at low (5 uM) and high (20 uM) contrasts. All DGE in the low contrast overlapped with the high contrast. No DEU genes overlapped between the gene-level contrasts except for *cyp35A1*. With the exception of *cyp35a1* in the DEU contrasts, the stipulation for inclusion to the table was that each family contained at least 2 unique genes within a contrast. The *let* and *unc* genes were alternatively spliced in different gene members across contrasts.

Category	Low Exposure (DGE)	Low Exposure (DEU)	High Exposure (DGE)	High Exposure (DEU)
Carboxylesterases (<i>cest</i>)	0	0	2	0
C-Type Lectin (<i>clec</i>)	2	0	3	0
Cytochrome P450 (<i>cyp</i>)	3	1	8	1
Immunity-Related GTPases (<i>irg</i>)	2	0	3	0
Glucuronosyl-transferases (<i>ugt</i>)	5	0	11	0
Lethal (<i>let</i>)	0	1	0	1
Uncoordinated structural proteins (<i>unc</i>)	0	2	0	2

In the linear model, 2 genes were detected that were related to cellular signalling: *ccr9a*, a membrane protein-coding gene which in humans is located in the thymus and intestines, and *rbpjl*, a transcriptional regulator of notch signalling. Other genes detected in the linear gene-level model were related to broad classes of enzymes:

aryl sulfatases (*arsj*), bone resorption proteases (*ctsk*), and zinc- and calcium-dependent proteases (*tll1*). *C1orf43*, which was detected in the linear DEU model, is a regulator of phagocytosis, but according to Ensembl has only 1 splice variant. Because DEU implicates alternative splicing and the production of multiple gene product isoforms, detecting genes with only 1 splice variant suggest these findings are either false positives or demonstrate that the *Gadus morhua* annotations lack all variants of *C1orf43*.

To evaluate how many alternatively spliced genes detected in this study are known to have only 1 splice variant, an assessment of the 44 unique genes detected in the DEU contrasts in *C. elegans* and their splice variants was performed by inspecting each gene's Ensembl webpage (Table 3). Of the 19 genes detected in the 5 uM DEU contrast, 6 (*let19*, *nhl2*, *rps18*, *cyp35A1*, *F30H5.3*, *W06H8.6*) are known to have only 1 splice variant. Of the 29 detected in the 20 uM DEU contrast, 11 (*cof2*, *daz1*, *moc2*, *paa1*, *unc54*, *hrde1*, *DH11.2*, *ddb1*, *B0238.9*, *ttbk8.2*, *Y27F2A.6*) are known to have only 1 splice variant.

To underscore if there was broad overlap between the biological functions of the detected genes, functional enrichment was performed. No clusters were enriched for DTU because a minimum of 3 genes is required for functional enrichment and DTU analysis yielded only 2 genes. The top clusters in the gene-level contrasts related to immune, xenobiotic, and pathogen responses (Table 4). No contrasts at the exon-level mapped significantly to a GO term. Many clusters were detected in the DGE and DEU linear models in *C. elegans*. The top clusters were metabolic, developmental, and reproductive processes. Only 1 KEGG pathway was enriched in the linear DGE model: metabolic pathways. In *G. morhua*, clusters related to a variety of regulatory processes, but the low number of DEGs meant that FDRs were higher than in *C. elegans*. The 1 uM gene-level contrast mapped to cellular responses to chemicals while the gene-level linear model mapped to broader metabolic, regulatory terms, similar to the *C. elegans* linear models.

Table 3: Summary of genes detected in *C. elegans* differential exon usage contrast models that contained only 1 splice variant according to Ensembl transcript tables. In parentheses are counts divided by total genes detected in the contrast or in both DEU contrasts.

DEU Contrast	1 Splice Variant Genes (% of total)
Low Exposure	6 (32)
High Exposure	11 (38)
Total	17 (39)

Table 4: Functional enrichment clustering of biological process GO terms and KEGG pathways from DAVID in differentially expressed genes and differential exon usage genes for *C. elegans* and *G. morhua*. Included beneath Gene Ontology terms and the Kyoto Encyclopedia of Genes and Genomes pathway are gene counts, fold enrichment (FE), and false discovery rates. Top enriched process in each cluster included, as calculated by Fisher's exact score. Minimum genes required for DAVID was 3. Marginal significance (0.073) reported in the 1 uM gene-level in *G. morhua* with FDR underlined.

Treatment (Total Clusters)	Cluster 1 (gene count; FE; p-value)	Cluster 2 (gene count; FE; p-value)	Cluster 3 (gene count; FE; p-value)	Top KEGG
<i>Caenorhabditis elegans</i>				
DGE: Low Exposure (2)	Innate Immune Response (6; 17.7; 1.3×10^{-4})	Xenobiotic metabolism (3; 71.3; 0.0032)		
DGE: High Exposure (4)	Cellular response to xenobiotic (7; 61.6; 5.2×10^{-8})	Defense response to other organism (9; 7.1; 2.5×10^{-4})	Glycolipid metabolism (3; 22.6; 0.037)	
DGE: Linear Model (72)	Defense response to other organism (50; 2.5; 1.2×10^{-6})	Amide metabolism (24; 3.3; 1.3×10^{-4})	Organic acid metabolism (50; 2.9; 5.6×10^{-8})	Metabolic pathways (77; 1.8; 2.2×10^{-8})
DEU: Linear Model (21)	Reproductive developmental processes (15; 4.5; 0.0021)	Positive regulation of development (12; 5.8; 0.0021)	Reproductive process (19; 3; 0.0069)	
<i>Gadus morhua</i>				

DGE: High Exposure (1)	Cellular response to chemical (3; 27.8; <u>0.073</u>)		
DGE: Linear Model (6)	Progesterone metabolism (3; 332; 0.0054)	Biological regulation (11; 2; 0.046)	Chemical response (5; 10.1; 0.019)

Discussion

Benzo[a]pyrene is a carcinogenic environmental pollutant that organisms respond to by altering their cellular transcriptional environment (Wang et al., 2010; Fang et al., 2013). Despite the well-understood role of differential gene expression in modulating plastic responses to benzo[a]pyrene (Wang et al., 2006), less is known about how alternative splicing may support stress responses. Published HTP RNAseq data from *C. elegans* and *G. morhua* PCLS exposed to varying concentrations of BaP was analyzed to evaluate the synergism of DGE and AS over a dose response gradient.

It is shown that most differentially expressed genes were not alternatively spliced in *Caenorhabditis elegans* and *Gadus morhua*. One gene, *cyp35a1*, exhibited differential expression and alternative splicing, suggesting that synergism only takes place in the most important genes responsible for metabolizing BaP (Figure 2). Or more broadly, plasticity was generated by DGE and AS independently of one another in most genes. This finding fits into previous results which showed that expression and splicing are mostly independent drivers of phenotypic plasticity, but that synergism occurs in fewer genes that are conserved for mitigating the effects of the primary stressor (Vaidya et al., 2025). In humans, researchers found that *in vitro* exposures of various chemicals known to cause oxidative stress to human alveolar adenocarcinoma cells found that AS occurred independently to DGE in stress response genes (Barutcu et al., 2024). Other investigators that profiled *CYP* gene expression of molluscs, in bullhead fish, and bovine epithelial cells found significant upregulation of *cyp1a* genes in response to BaP exposure (Li et al., 2021; Williams & Hubbersty, 2014; Bourque et al., 2018), but did not evaluate AS. However, when researchers exposed zebrafish embryos to BaP from fertilization to 96 hours post-fertilization, gene-level upregulation and alternative splicing took place in *CYP* genes, as in this study (Fang et al., 2015). Too few studies have examined if AS is

synergistic to DGE in response to other chemical stressors to be able to conclude that AS is a fundamental part of chemical stress responses. However, these results and those reported elsewhere suggest that gene-level upregulation of *CYP* genes in response to BaP exposure is widespread in the animal kingdom, and that when researchers test for AS in exposure studies, AS is conserved as a synergistic mechanism to DGE that generates plasticity.

Second, this study showed that rates of DGE and AS increased as dose increased in *C. elegans*, but that there was little evidence of AS in *G. morhua*. This was possibly due to lower concentrations of BaP and therefore lower cellular stress. Other investigators similarly showed that dose dependent rates of alternative splicing mirror gene expression at higher temperatures in *Drosophila melanogaster*, suggesting that alternative splicing may switch from being an independent to a synergistic mechanism at an inflection point (Jakšić & Schlötterer, 2016). However, point of departure (POD) was not specifically tested in this study.

Finally, it is shown that using exposure as a linear predictor of DGE and AS detects genes that were differentially expressed and alternatively spliced at low-magnitudes, but possibly contain more false positives. In future experiments, it may be prudent to either increase the breadth of experimental groups to increase strength and confidence in linear predictions, as 3 data points are likely inadequate to estimate a relationship that is likely complex. Alternatively, a linear model may not be suitable to predict dose response relationships at all. This could be true if either the stressor exhibits non-linear effects on expression or splicing, if individual gene products differ in their response curves, if there is a point of departure where cytotoxicity occurs, or, most confoundingly, if induction of one mechanism affects the other mechanism interactively, as has been previously suggested (Yang et al., 2016).

A more rigorous protocol was used in the *C. elegans* analysis, and different software in the *G. morhua* analysis, but these findings are mostly concordant with the original investigators (Abbass et al., 2021; Yadetie et al., 2018). While the *C. elegans* study investigators found 374 DEGs in the 5 and 20 uM contrasts, 73 DEGs were detected in the same contrasts in this study. An explanation for this difference may be that the original investigators used reads per kilo base (RPKM) counts to improve the

comparability when sequencing depth, gene length, and expression levels differ. However, RPKM is not an adequate normalization method ahead of gene expression analysis (Zhao et al., 2020). Despite quantitative differences, many identical genes were detected within the *cyp* and *ugt* superfamilies. The *G. morhua* study investigators detected 1 gene in the 0.01 uM contrast and 13 in the 1 uM contrast. This analysis produced a similar number of DEGs and 4 of the genes detected in the 1 uM contrast were detected by the original investigators in the same contrast. However, this study included the effect of individual as a covariate in the DGE analysis which the original investigators did not.

Several gene superfamilies were overrepresented in *C. elegans* DEGs: C-Type lectins, glucuronosyltransferases, cytochrome P450s, and immunity-related GTPases (Table 2). These genes were clustered to xenobiotic metabolism, immune, and defense response GO terms (Table 4). *CYPs* and *UGTs* are known to modify the solubility of xenobiotics and enhance their excretion. Drugs such as primaquine and lansoprazole, and pesticides such as atrazine and dichlorvos are known to activate *cyp35A1* in *C. elegans*, demonstrating the broad function of this gene family in xenobiotic metabolism (Largiot et al., 2022). Similarly, C-Type lectins are known to recognize a variety of ligands during immune responses (Brown et al., 2018). Immunity-related GTPases are known to bind to pathogenic lipid membranes during immune responses too (Taylor, 2018).

The DEGs detected in *G. morhua* were enriched for similar functions to those in *C. elegans*, although they were fewer in number and less diverse. This may have been due to several reasons. Sequencing only took place on one fish's PCLSs within a replicate. Each replicate only accounted for 1 individual's biological variation, which contrasts with the *C. elegans* study where thousands of worms were homogenized per RNA extraction. Secondly, the concentrations used (0.01 & 1 uM BaP) may not have been high enough to elicit a transcriptomic response from the liver slices. While this choice by the original investigators was likely strategic to evaluate the effect of low, environmental concentrations of BaP, these results suggest that the concentration used was too low to evaluate acute response mechanisms, especially when comparing with higher concentrations in *C. elegans*. A limitation of functional enrichment is the statistical requirement of 3 or more genes to define a term or

pathway, but this may exclude effects of one or two genes that are important in responding to stress, especially if no similar genes are induced (Euling et al., 2013). Finally, the exon-level *G. morhua* models did not consider variation that stemmed from individual, which was greater than inter-treatment variation. This was omitted because models were too complex to run locally. It is uncertain whether the approach used in the exon-level *G. morhua* models was able to tease the effect of benzo[a]pyrene exposure on transcription from noise and individual fish variation (Figure 1B). Therefore, it was not possible to confidently test all 3 aims of this study in *G. morhua* because very few DEGs were detected, only 1 alternatively spliced gene was detected (Table 1), and the DEU models did not account for the majority of sample variation.

By using both DTU and DEU as analogs of alternative splicing, this study demonstrates the benefits and limitations of employing both metrics. While transcript usage is a direct measure of unique, mature gene products, these analyses captured very little DTU. This may have been because the annotations used did not contain all splice variants, especially in *G. morhua*, a non-model organism, and that accuracy varies according to the annotation source (Chisanga et al., 2022). Conversely, DEU is an indirect measure of AS. When the ratios of exons from a gene change, AS is implicated because DGE does not change the transcriptional ratio of exons within a gene. However, the genome annotations used in this study may have lacked detail about all transcript isoforms in *C. elegans* too. 39% of genes detected in *C. elegans*' DEU contrasts are known to have only 1 splice variant (Table 3). Although DTU is a stronger, more direct measure of AS, these results demonstrate that restraints in bioinformatic resources may limit its applications in ecological research. Future research may benefit from using RT-PCR to validate findings from HTP-RNAseq data in select genes that are hypothesized to be alternatively spliced to increase confidence in results.

This study did not systematically evaluate which genes AS was detected in if they did not intersect with differentially expressed genes. However, several genes where AS took place were linked to muscle or developmental processes (Table 4). This supports previous hypotheses stating that AS is enriched in genes related to development (Bush et al., 2017; Jacobs & Elmer, 2021). AS was also previously

detected in muscle tissue in response to cold stress (Healy & Schulte, 2019). However, this was not detected in *G. morhua*, which is likely because RNA came only from liver slices, not muscle tissue.

Future investigation into the fundamental mechanisms of regulating ecotoxicological stress responses can benefit from integrating more perspectives than were used in this perspective. This may be done to improve the understanding of multiple scales of xenobiotic action and therefore gain a holistic understanding of the toxicant. For example, an integrated toxicological approach that seeks to benchmark no observed adverse effect levels, POD, and LC50s of DGE and AS could enhance the reproducibility and translation of the research to support regulatory processes (Barutcu et al., 2024). Similarly, by integrating histology and other analyses such as proteomics and metabolomics, crosstalks between immediate transcriptional regulatory networks and large-scale effects on metabolism could be elucidated, as was done by researchers investigating cold responses in Tibetan teleost fish (Liu et al., 2023). Finally, although this study attempted to assess the relationship between DGE and AS in organisms with different genes responsible for xenobiotic metabolism, future research that tailors an experimental design to better suit this question could evaluate an exciting new hypothesis that bridges human toxicology and ecotoxicology: that boundaries in toxicity may lie in homology, and that toxicity is inherited (Colbourne et al., 2022).

It is demonstrated that alternative splicing plays a predominantly independent role to differential gene expression in response to benzo[a]pyrene in *C. elegans*, but that there was conservation of alternative splicing in 1 key xenobiotic metabolism gene. Rates of alternative splicing increase alongside differential gene expression along a dose gradient. Lastly, using chemical exposure as a linear predictor may detect low-magnitude expression and splicing, but care must be taken in the experimental design to establish where these fundamental mechanisms diverge from linear patterns.

Acknowledgements

The author acknowledges and thanks the support of Drs. Marius Wenzel and Martin Barker for their technical and pastoral support during the planning, execution, and writing of this study.

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Data & Code Accessibility

HTP RNAseq data are publicly available. NCBI Bioproject accession codes are PRJNA638820 and PRJNA418681 for *Caenorhabditis elegans* and *Gadus morhua*, respectively. Code used for analyses are available on Github:
<https://github.com/danielheltberg/BaPTranscriptomics>.